into a dynamic, structured, and intelligently managed mobility layer, complementing ground transportation and reshaping urban movement patterns.

9.4.1 The rise of passenger drones for urban transportation

The emergence of passenger drones, commonly referred to as eVTOL (electric vertical take-off and landing) air taxis, marks a significant step toward redefining how people move within and between urban environments. These aircraft are engineered to take off and land vertically, eliminating the need for runways and enabling operations from compact vertiports located on rooftops, transit hubs, commercial complexes, or dedicated platforms. Driven by advancements in electric propulsion, lightweight composite materials, autonomous flight systems, and smart navigation sensors, passenger drones aim to provide fast, direct aerial travel over congested road networks, thereby reducing commute time and easing pressure on urban transportation systems.

Several companies such as Volocopter, Joby Aviation, EHang, Lilium, and Airbus have developed and tested full-scale passenger drone prototypes capable of transporting one or more passengers safely and efficiently over short-to-medium intra-city distances. Pilot trials in cities like Dubai, Singapore, Paris, and select US regions demonstrate growing readiness for operational deployment. These passenger drones employ redundant safety systems, distributed electric propulsion units, autonomous stabilization controls, and low-noise rotor configurations, making them suitable for dense residential and commercial environments.

As passenger drones progress toward commercialization, they are increasingly viewed as a central component of urban air mobility (UAM) networks. Their integration promises to introduce rapid, low-emission, point-to-point air travel, particularly for airport transfers, medical